

Batik Classification using MobileNetV3 and YOLOv11 Classification

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Abstract— Batik is a profound representation of Indonesian cultural heritage, recognized by UNESCO, characterized by a vast diversity of motifs unique to various regions[1]. This research aims to develop and compare automated image recognition models to classify these intricate patterns using Deep Learning techniques. Specifically, this study evaluates the performance of MobileNetV3-Large, a lightweight Convolutional Neural Network (CNN) optimized for mobile devices, against YOLOv11-Medium, a state-of-the-art real-time object detection and classification architecture. The models were trained and validated using a dataset of diverse Batik classes with a 70/20/10 split. The study utilizes metrics such as accuracy, F1-score, training time, and confusion matrices to assess performance. Results indicate that YOLOv11-Medium outperforms MobileNetV3-Large with a Test Accuracy of 83.1% compared to 78.1%, while also requiring significantly less training time.

I. INTRODUCTION

Indonesian Batik is a traditional art form that holds significant cultural value. The diversity of Batik motifs, which vary from the "Megamendung" of West Java to the "Cendrawasih" of Papua, reflects the rich traditions of the archipelago. In the era of Society 5.0, preserving this heritage through digital archiving and automated recognition systems is essential. However, the visual complexity and similarity between different Batik patterns pose a significant challenge for traditional image processing methods.

This report investigates the following questions:

1. How do lightweight (MobileNetV3) and specialized detection (YOLOv11-cl) architectures compare on fine-grained Batik classification?
2. What are the computational efficiency tradeoffs between training time and accuracy?
3. Which classes present persistent classification challenges, and what are the underlying causes?
4. What architectural features contribute to superior performance?

II. LITERATURE REVIEW

Prior work has established CNN-based approaches as the standard for Batik pattern recognition[2][3][4]. Perdana et al. (2025) demonstrated that transfer learning with pre-trained models (DenseNet169, MobileNetV2, ResNet50, DenseNet201, InceptionV3) achieves testing accuracy between 69-91% on 3-class Batik datasets[4]. Their study emphasized the importance of data augmentation and custom classification layers in improving generalization.

Arifin & Nurfaizah (2024) developed a CNN-based multi-class classifier achieving 93.18% training accuracy on a diverse Batik dataset, highlighting the effectiveness of proper preprocessing, normalization, and augmentation strategies[2].

Transfer learning leverages knowledge acquired from large-scale pre-training (typically ImageNet) to accelerate convergence and improve accuracy on domain-specific tasks[5]. Two primary strategies emerge:

Feature Extraction: Freezing early layers and training only final layers, reducing computational cost.

Fine-tuning: Selectively unfreezing layers to adapt pre-trained weights to the target domain.

MobileNetV3 was specifically designed for mobile and edge deployment, employing depthwise separable convolutions and neural architecture search (NAS) to optimize the accuracy-latency frontier[6].

The latest YOLO release (v11) extends the object detection framework to classification tasks, offering a unified architecture across detection and classification with refined training strategies, improved data handling, and optimized inference pipelines[7]. YOLOv11-Medium serves as a balance between speed and accuracy, retaining multi-scale feature extraction from the detection backbone while applying classification heads.

III. METHOD

A. Dataset Preparation

The dataset comprises 2,330 images across 30 distinct Indonesian Batik motif classes which are sourced from the Hugging Face dataset repository by Muhammad Salman[8]. Which is then further modified in Roboflow to remove classes with insufficient dataset (<30 images), This dataset provides a comprehensive benchmark for multi-class Batik pattern recognition, supporting research in both computer vision and cultural heritage preservation. Images were organized into stratified 70% / 20% / 10% train/validation/test splits to preserve per-class proportions across subsets. The 30 classes represent major regional Batik traditions spanning Indonesia's geographic and cultural diversity, from Aceh in the north to regions across Kalimantan, Papua, Java, Sumatra, Sulawesi, and the Nusa Tenggara islands.

Batik Motif	Total	Train	Valid	Test
Parang	207	148	38	21
Jawa Barat Megamendung	202	143	37	22
Kalimantan Dayak	196	132	41	23
Papua Cendrawasih	128	95	24	9
Yogyakarta Kawung	125	84	28	13
Sumatera Utara Boraspati	80	54	19	7
Sumatera Barat Rumah Minang	80	54	17	9
NTB Lumbung	80	58	11	11
Sulawesi Selatan Lontara	80	53	19	8
Lampung Gajah	80	58	14	8
Maluku Pala	79	55	15	9
Madura Mataketeran	78	46	22	10
Papua Tifa	78	50	23	5
Papua Asmat	78	58	11	9
Corak Insang	75	53	13	9
Sidoluhur	50	33	10	7
Tambal	50	34	12	4
Sogan	50	33	15	2
Lasem	50	33	10	7
Keraton	49	32	15	2
Pekalongan	49	32	11	6
Priangan	47	34	9	4
Sekar	47	34	8	5
Ikat Celup	46	31	13	2
Bali Barong	43	31	8	4
Jakarta Ondel Ondel	43	36	4	3
Sidomukti	42	28	10	4
Jawa Timur Pring	40	30	5	5
Aceh	40	32	3	5
Bali Merak	38	30	4	4
Total	2,330	1,624	469	237

Table 1. Complete dataset composition across all 30 Indonesian Batik motif classes sourced from Hugging Face (muhammadsalmanalfaridzi/Batik-Indonesia) and was then further modified in Roboflow. Classes are ranked by total dataset size, reflecting both documentation prevalence and artisanal tradition popularity.

B. Train/Validation/Test Split

Images were divided using a 70% / 20% / 10% stratified split to maintain class proportions across subsets, ensuring representative evaluation across all motifs:

Split	Images	Percentage
Training	1,624	69.70%
Validation	469	20.13%
Testing	237	10.17%

Table 2. Train/Validation/Test split for the dataset used.

C. Preprocessing & Augmentation

Both models received images resized to 224×224 pixels. YOLOv11 scales inputs to $[0, 1]$ internally, while MobileNetV3-Large performs its own rescaling inside the model, since the Keras mobilenet_v3 preprocess_input function is a pass-through. Augmentation strategies are made to be as similar as possible to reflect each framework's design philosophy:

- YOLOv11 Augmentation (via Ultralytics):
- Rotation: ± 15 degrees
- Horizontal flip: 50% probability
- Mosaic augmentation: 100% (default for YOLO, creates synthetic multi-image compositions)
- Scale/zoom: 0.2 range
- Erasing: 0.2 probability
- HSV perturbation: $H \pm 0.01$, $S \pm 0.1$, $V \pm 0.1$

MobileNetV3 Augmentation (via ImageDataGenerator + custom):

- Rotation: ± 15 degrees
- Zoom: 0.2 range
- Width/height shift: $\pm 10\%$
- Horizontal flip: 50% probability
- Random erasing: 0.2 probability (custom hook)
- Normalization: handled inside the MobileNetV3 model (preprocess_input is a no-op)

D. Model Architectures

a. YOLOv11-Medium Classification

YOLOv11-Medium (yolo11m-cls.pt) is a medium variant of the YOLO family optimized for classification tasks[7]:

- Backbone: Multi-scale feature extraction with C3k2 blocks (convolutional blocks with kernel grouping)
- Total parameters: 10,391,646
- Input size: $224 \times 224 \times 3$
- Output: Softmax probabilities across 30 classes
- Training configuration:
 - Batch size: 32
 - Optimizer: AdamW (automatic LR selection via auto mode)

- Epochs: up to 30 with early stopping (patience=10)
 - Learning rate: auto-determined
- b. MobileNetV3-Large (ImageNet pre-trained)
- MobileNetV3-Large (ImageNet pre-trained) backbone combined with custom classification head[6]:
- Backbone: MobileNetV3-Large (ImageNet weights)
 - Frozen layers: Backbone layers except final 30 (allows selective fine-tuning)
 - Custom head architecture:
 - Global Average Pooling 2D over the final backbone feature map
 - Dropout: 0.3 (regularization to prevent overfitting)
 - Output layer: 30 units, softmax
 - Training configuration:
 - Batch size: 32
 - Optimizer: Adam ($lr=1e-4$)
 - Epochs: up to 30 with early stopping (patience=10)
 - Loss: Categorical cross-entropy

IV. RESULT AND DISCUSSION

A. Accuracy, Loss, and Confusion Matrixes Results

Both models were trained on identical hardware (Tesla T4 GPU, ~15GB VRAM) using the same dataset paths and evaluation protocols, enabling a fair architectural comparison unconfounded by infrastructure or data handling differences.

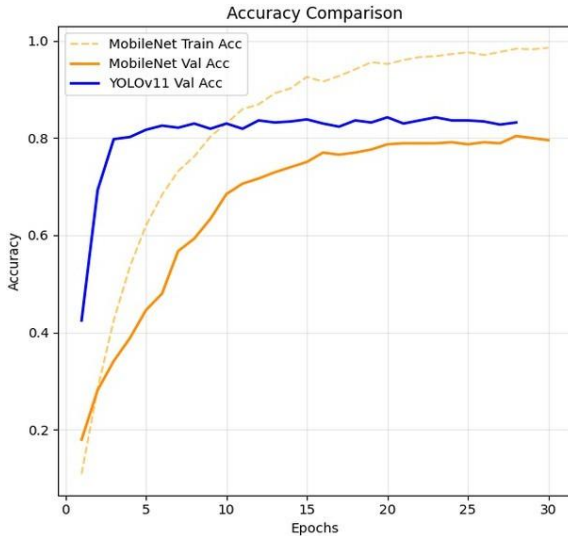


Figure 1. Accuracy comparison between YOLOv11 and MobileNetV3

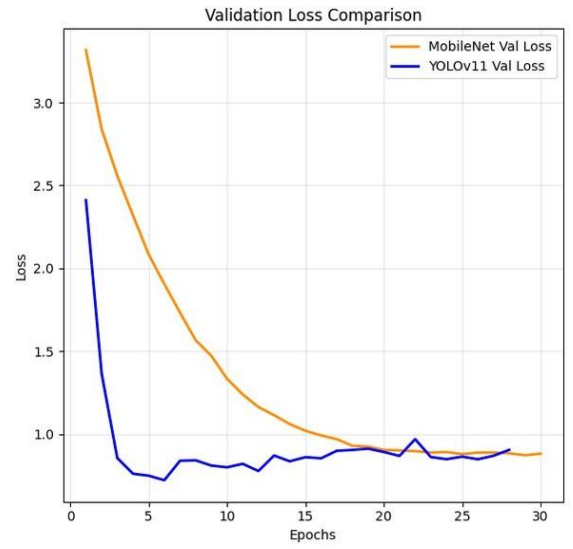


Figure 2. Validation Loss comparison between YOLOv11 and MobileNetV3

YOLOv11 reaches its best validation accuracy relatively early: in the run log, the best performance is reported at epoch 18, after which early stopping (patience 10) halts training at epoch 28, with validation top-1 accuracy in the ~83–84% range around that best epoch. The improvement is rapid in the first few epochs (validation top-1 jumps strongly within roughly the first 5–6 epochs), and the curve is comparatively smooth, which is consistent with stable optimization during training.

MobileNetV3 improves more gradually across the full training schedule and does not trigger early stopping, reaching a final validation accuracy of about 0.7953 at epoch 30 in the training log. Compared to YOLOv11’s earlier peak, this suggests slower feature adaptation in this setup (backbone fine-tuning plus the chosen GAP to Dropout to Dense head).

Looking at the validation losses, the YOLOv11 run shows loss settling into a relatively stable band after the early/mid epochs, while MobileNetV3’s validation loss stays higher for longer and only trends down toward the end of training (e.g., around ~0.88 by epoch 30).

Metric	YOLOv11-Medium	MobileNet V3-Large	Delta
Test Accuracy	0.8312 (83.1%)	0.7806 (78.1%)	+5.1 PP
Weighted F1-Score	0.8286	0.7734	+0.055
Macro F1-Score	0.7340	0.6770	+0.057
Training Time (min)	6.69	12.12	-5.4 min
Training Time (sec)	401.2	727.5	-44.9%

Table 3. Final test-set metrics on 237 held-out images. YOLOv11 achieves both higher accuracy and lower computational cost, making it the superior baseline for deployment and future research.

The 5.1 percentage-point accuracy advantage for YOLOv11, combined with 45% faster training, makes YOLOv11-Medium the superior baseline for this dataset and task. The weighted F1-score advantage

(0.055 absolute difference) reflects better handling of imbalanced classes, while the macro F1 improvement (0.057) demonstrates more robust per-class generalization despite class imbalance.

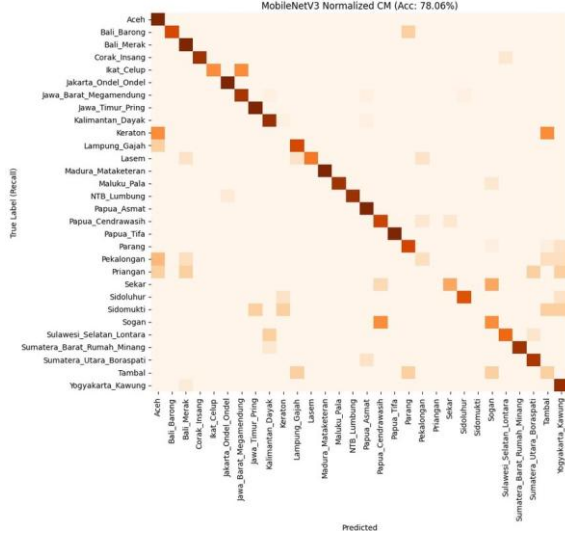


Figure 3. Normalized Confusion Matrix of MobileNetV3

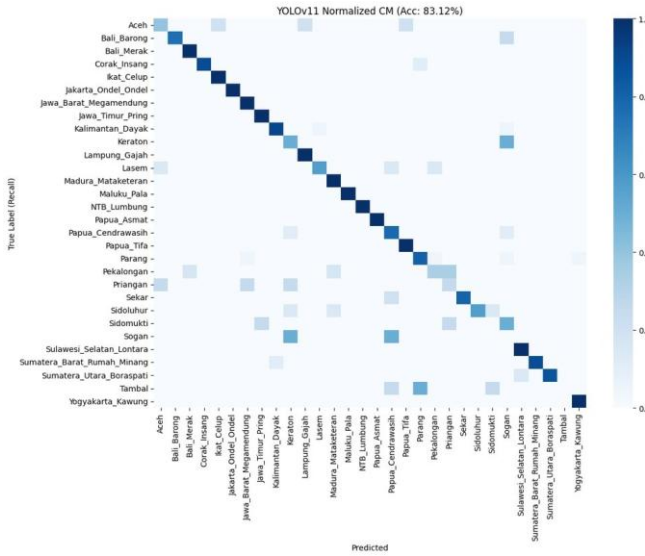


Figure 4. Normalized Confusion Matrix of YOLOv11 Classification

The normalized confusion matrices visualize per-class correct-prediction rates (i.e., recall on each true class), where stronger diagonal cells indicate higher recall for that class and lighter off-diagonal cells indicate fewer misclassifications into other motifs. In this comparison, YOLOv11 (Figure 4) shows a more concentrated diagonal pattern than MobileNetV3 (Figure 3), suggesting more reliable per-class predictions and fewer confusions among visually similar Batik motifs.

Both models still display a clear diagonal structure overall, which confirms that they learned meaningful, class-specific visual patterns. However, YOLOv11’s matrix exhibits higher diagonal intensity (more consistent recall across the 30 classes) and fewer prominent off-diagonal regions (fewer systematic confusions), which aligns with its stronger class-level generalization. The off-diagonal patterns that do

appear tend to be relatively symmetric, implying that most remaining errors are driven by genuine visual similarity between motifs rather than artifacts or one-sided bias in the predictions.

B. Model Performance

Several architectural and training factors explain YOLOv11’s consistent advantage over MobileNetV3 in this Batik motif task. YOLOv11’s backbone benefits from a multi-scale feature hierarchy, combining representations at multiple receptive field sizes (for example, $8\times$, $16\times$, and $32\times$ spatial reduction), which helps it capture both fine-grained textures and larger geometric structures within motifs. In contrast, MobileNetV3 is optimized for efficient single-stream classification and can lose some contextual cues that matter when separating visually similar regional patterns. YOLOv11 also tends to converge faster because its training pipeline and augmentation strategy (including mosaic-style compositions) expose the model to more varied effective training examples per epoch, helping it navigate the loss landscape more efficiently. MobileNetV3’s slower improvement can indicate that the chosen head design and fine-tuning schedule are not perfectly matched to the dataset, and that the backbone may require a different adaptation strategy to generalize as well.

Finally, YOLOv11 benefits from implicit regularization produced by its augmentation schedule, which can reduce overfitting without relying heavily on dropout. At the same time, YOLOv11-Medium achieves a strong balance between depth and width for a 30-class setting, maintaining expressive capacity at a competitive parameter count, while MobileNetV3’s depthwise separable design trades representational richness for latency, which can be limiting for fine-grained textile discrimination.

Fine-grained motif recognition remains challenging because some classes share repeated geometric elements, crossed lines, and similar color palettes, making pixel-level separation difficult. For these cases, improvements can focus on mechanisms that emphasize discriminative regions, such as attention modules, so the model learns to rely on motif-specific details like borders, corners, and subtle gradients.

Beyond attention, metric learning approaches can be used to learn embeddings that increase inter-class separation and reduce intra-class variance using objectives such as triplet loss or contrastive learning. Additional gains may come from saliency-guided augmentation, where interpretability maps to identify confusing regions and augmentation is targeted to strengthen decision boundaries in those areas, and from hierarchical classification where motifs are grouped into coarse families before predicting the fine-grained class.

Both models achieve good test accuracy, but their training to validation behavior suggests different generalization characteristics. YOLOv11 shows relatively stable training and validation accuracy (around 83% for both), which is consistent with low

overfitting and more robust feature learning. MobileNetV3 shows a large gap, with very high training accuracy (around 98%) but notably lower validation accuracy (around 79.5%), which is a common sign that the model is memorizing training examples rather than learning transferable features.

To verify and strengthen generalization, it is useful to run multi-fold cross-validation to measure performance variance across splits, evaluate on a truly independent external Batik collection to test domain robustness, and analyze confidence calibration to detect overconfidence or underconfidence by class. Feature visualization methods can also help reveal what each model is focusing on, making it easier to design better augmentation policies or revise the architecture.

V. CONCLUSION

This comparative study provides strong evidence that YOLOv11-Medium is the better baseline for Indonesian Batik motif classification on the evaluated dataset. On the held-out test set, YOLOv11-Medium achieved 83.1% accuracy, a weighted F1-score of 0.829, and a macro F1-score of 0.734, outperforming MobileNetV3-Large by 5.1 percentage points in accuracy, 0.055 in weighted F1, and 0.057 in macro F1. The macro F1 advantage is especially meaningful because it reflects more consistent performance across classes, rather than only improving results on the most frequent motifs. Overall, these metrics indicate that YOLOv11-Medium delivers stronger and more balanced recognition quality for the full 30-class Batik problem under the same evaluation protocol.

Efficiency results also favor YOLOv11-Medium and strengthen its practical value. YOLOv11 completed training in 6.69 minutes compared to 12.12 minutes for MobileNetV3-Large, corresponding to a 45% reduction in training time while still achieving higher test-set performance. This matters because faster training reduces experimentation cost, makes iterative refinement more feasible, and lowers compute overhead for teams that need to rerun training under consistent pipelines. In short, YOLOv11-Medium provides a better accuracy to time tradeoff in this study, which is an important criterion when models must be trained, validated, and maintained efficiently.

From a cultural heritage perspective, automated Batik classification can accelerate documentation and preservation by making motif identification faster and more scalable than manual labeling alone. With over 83% test accuracy and low training overhead, YOLOv11-Medium is suitable for deployment contexts that benefit from real-time or near-real-time

predictions, including edge environments such as mobile phones, museum kiosks, and tools used in artisan workshops. In these settings, a reliable classifier can help artisanal communities by supporting knowledge transfer, help educators build interactive learning tools that link patterns to regional context, support museums in cataloging and preliminary authentication workflows, and assist e-commerce verification efforts that aim to reduce counterfeit risks.

At the same time, the results also indicate that performance is not uniform across all motifs, since fine-grained similarity and limited examples for certain classes can still produce confusions at the class level. This means the system is best framed as a decision-support tool that can rapidly propose likely motif labels, while still benefiting from expert oversight for ambiguous cases, especially in situations where the cost of a misclassification is high. Even with these limitations, the overall outcomes show that YOLOv11-Medium provides a strong, efficient baseline for Batik motif recognition and a practical foundation for real-world usage in cultural preservation workflows

VI. REFERENCES

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